

The Mass of an Object and its Effect on Damping in Damped SHM

1. Research Design

1.1. Introduction

When solving problems involving waves and harmonic motion, it was common practice to neglect resistive forces such as air or friction to solve for systems with simple harmonic motion: However, real-world systems are influenced heavily by damping forces which play an essential role in determining the behavior of oscillatory motion.

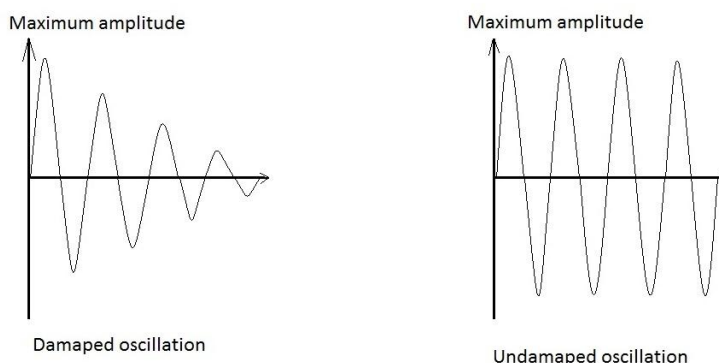


Figure 1: Position-time graph of Damped (left) and Undamped (right) harmonic motion

Wanting to explore the effect of resistive forces that affect oscillation, I began to explore the fundamentals that govern a damped harmonic system, examining how different factors affect the damping effect of an oscillation system. This paper, in particular, focuses on the mass in a harmonic system, mainly because it was perhaps the most interesting, but quantifiable, factor that affected the system.

1.2. Purpose

Therefore the purpose of this study is to examine the following question: What is the effect of increasing mass (40g, 60g, 80g, 100g, 120g, 140g) on the damping effect of the system, quantified by the damping ratio of a mass-spring damped harmonic motion system, measured with an ultrasonic motion sensor?

1.3. Background

In order to understand and draw a relationship between mass and damping ratio, it is first nessecary to understand DHM systems. Essentially, they are similar to SHMs but with the added consideration of resistive forces, resulting in a damping where mechanical energy is lost (Morin 2008). Due to the small scale of this experiment, it will primarily focus on understanding **underdamped systems**, which are systems that oscillate back and forth m times until they reach equilibrium (Garg 2009). Below is the free body diagram (FBD) for such a system with the velocity (v) being that of the object in motion:

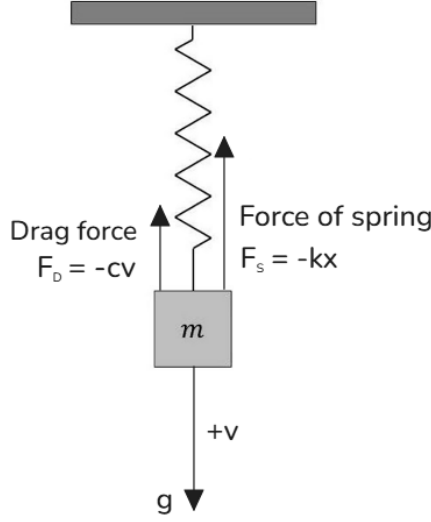


Figure 2: FBD diagram of a mass-spring system (Excalidraw)

Using the FBD diagram shown in Figure 2 and the general understanding of a damped harmonic system, the net force equation for a damped mass-spring system can be described as such:

$$\begin{aligned}
 F_{net} &= F_D + F_s \\
 ma &= -cv - kx \\
 ma + cv + kx &= 0
 \end{aligned} \tag{1}$$

where:

- c – damping coefficient (kg/s)
- v – velocity (m/s)
- x – net displacement of spring (m)
- k – spring coefficient (kg/s²)

The damping coefficient (c) quantifies the rate at which energy within a system is dissipated by resistive forces, essentially measuring the strength of the resistive forces (Morin 2008).

This equation can be solved by re-writing it as a 2^{nd} order derivative by re-writing all the terms with respect to time like so:

$$m \frac{d^2x}{dt^2} + c \frac{dx}{dt} + kx = 0$$

Solving this equation yields us the equation for the instantaneous position of a DHM system solution for this was found to be the following (Morin 2008):

$$x(t) = Ae^{-\zeta\omega_0 t} \cos(\omega t + \phi) \quad (2)$$

Where:

$x(t)$ – instantaneous position time t (m)

A – initial amplitude (m)

ζ – damping ratio (no units)

ω_0 – undamped angular freq. of oscillation (radians / s)

ω – damped angular freq. of oscillation (radians / s)

ϕ – phase angle at $t = 0$ (radians)

While the equation tells us the relationship between the position and the damping ratio, it does not isolate the effect of the *mass* on the damping ratio. To do so, the damping ratio needs to be examined further. The damping ratio (ζ) measures how quickly a system returns to equilibrium compared to a critically damped system (Morin 2008) and can be defined as such:

$$\zeta = \frac{\text{actual damping}}{\text{critical damping}}$$

The actual damping of a system is described by the damping coefficient which quantifies the absolute strength of the damping effect. The critical damping of a system has been derived to be equal to $2\sqrt{km}$ (Franklin 2020).¹ With this, equation (1.3) can be re-written as:

$$\zeta = \frac{c}{c_c} = \frac{c}{2\sqrt{km}} \quad (3)$$

A clear relationship between mass and the damping ratio can be established, but measuring the damping ratio is challenging due to the difficulty of determining the damping coefficient experimentally, as it depends on material properties and its resistance to motion (Garg 2009). Another way to look at equation (2) is to examine the exponential term as it helps describe the decay of the motion towards equilibrium.

¹ Note that when $\zeta < 1$, the system is underdamped, when $\zeta = 1$ the system is critically damped, and when $\zeta > 1$, the system is overdamped (Morin 2008).

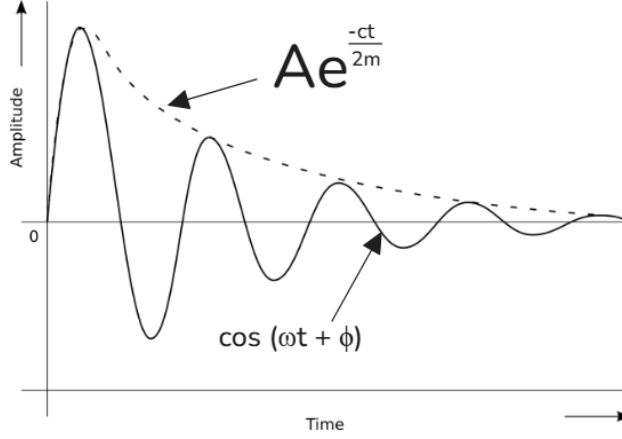


Figure 3: Graph of damped harmonic motion

Examining Figure 2, the graph of the DHM system is enveloped by the exponential decay component ($Ae^{-\frac{ct}{2m}}$). Thus, to measure the rate at which damping happens, the decay ratios of the peaks can be examined (as seen in Figure 3):

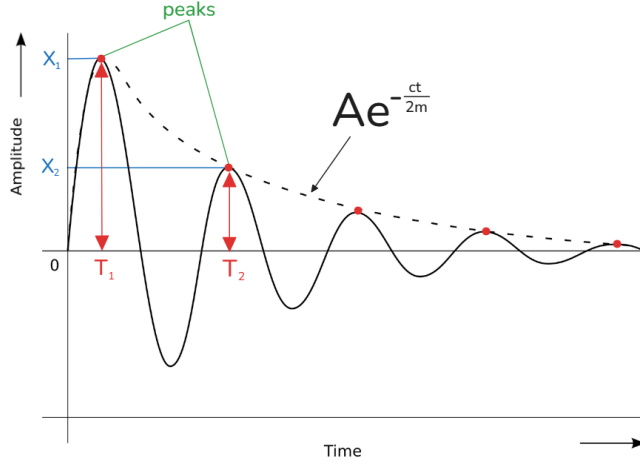


Figure 4: Graph of damped harmonic motion with peaks at X_1 and X_2

To describe the damping effect seen between the two successive peaks (X_1 and X_2) as seen in Figure 3 to describe the exponential decay, we can use the **logarithmic decrement** (represented by δ) which is represented by the following (University 2001):

$$\delta = \ln \frac{X_1}{X_2} = \ln \frac{x(t)}{x(t+T)} \quad (4)$$

where:

$x(t)$ – (initial) amplitude at time t (m), X_1

$x(t + T)$ – amplitude at time t after 1 cycle/oscillation (m), X_2

To find the damping ratio using the logarithmic decrement, the following equation can be used (University 2001):

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$

However, in the case of a lightly damped DHM system (where the system is well below critical damping and $\zeta \ll 1$, (University 2001)) we can assume that $\delta^2 \rightarrow 0$, and as such the expression above can be simplified to the following approximation:

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}} \approx \frac{\delta}{\sqrt{4\pi^2}} = \frac{\delta}{2\pi} \quad (5)$$

Thus, to find the damping ratio for any mass, we can use the logarithmic decrement from a position-time graph as demonstrated by equation (5).

1.4. Hypothesis

To determine the relationship between the mass and the damping ratio, we can linearize equation (3) as such:

$$\begin{aligned} \zeta &= \frac{c}{2\sqrt{km}} \\ \frac{1}{\zeta} &= \frac{2\sqrt{km}}{c} \\ \frac{1}{\zeta^2} &= \left(\frac{4k}{c}\right)m \end{aligned} \quad (6)$$

Through this, it becomes clear that as the mass of the system is increased the damping ratio will decrease proportionally, as $m \propto \frac{1}{\zeta^2}$. Thus it is hypothesized that as the increments of mass increase, the damping effect in the DHM system will be less pronounced/decrease as the damping ratio decreases. The slope of such a relationship is expected to be equal to $(4k/c)$, with the y-intercept being 0 for $m = 0$.

1.5. Experimental Variables

Independent variable (IV): The independent variable for this investigation is the mass (kg) of the mass-spring system, starting at 40g and increasing in increments of 20g: 40g, 60g, 80g, 100g, 120g, and 140g. These masses were selected through trial and error to produce noticeable damping while being measurable by the sensor, taking into consideration the experiment's limitations. Ideally, any mass with even increments could be used if feasible.

Dependent Variable (DV): The dependent variable for this investigation is the damping ratio (no units), which provides a way to quantify the damping effect of a given system. To calculate it, the logarithmic decrement will be used. To determine the accuracy, an approximation of the slope involving the spring constant and damping coefficient will be approximated will be used.

Control variables	Rationale	Method of control
Spring	Changing the spring means a change in the spring constant, which can affect the damping of the system.	The same spring was used throughout all the experiments, with the same spring constant.
Applied force on the system	Additional force on the system would add another variable to the DHM system described in equation (1), thus changing the experiment entirely.	The spring will be compressed to its original state, and (without applying any external forces), let go vertically.
Damping coefficient	Changing the aerodynamics of the mass-hanger spring system could impact the resistive forces of the system, thus varying the damping coefficient.	Since motion occurs in 2D (up and down) the cross-sectional area was kept constant by using slotted weights matching the base of the mass hanger.
Duration of trial (t)	Varying the duration of each trial would not allow for a proper analysis of the damping effect, as during different points of time, the amplitude is different.	The ultrasonic motion sensor will be set to record the measurement for 120 seconds.
Slotted weights	Changing the shape and type of slotted weights could change the damping coefficient or mass of the system.	The same type of slotted mass is used.
Tools of measurement	Using different ultrasonic sensors may yield different results due to their condition and ability to record height.	The same ultrasonic motion sensor was used throughout all the trials.
Drag forces	Air currents and wind could impact the drag forces involved in the resistance of the experiment.	Experiment was conducted in a controlled environment, away air ducts and other sources of wind.

Table 1: Table of control variables

1.6. Materials and Apparatus

Materials/Tools	Dimensions/Requirements
Weight Hanger (x1)	Chromium mass-hanger with a hole instead of a hooked end. This was chosen since hooked end weight hangers were less accurate since they would induce a faint "rocking" 3D motion which would interfere with the sensor reading.
Slotted Weights (x7)	20g Brass plated slotted weights with $\pm 2\%$ (as listed by manufacturer). These were chosen due to their ease of access as well as low relative uncertainty in comparison to other alternatives such as Platinum ones.
Spring (x1)	Small metal spring with a hooked end with a spring constant of 5886 g/s.*
Ultrasonic Sensor (x1)	Vernier Go!Motion; must be able to record and tabulate the position of the oscillations into the Vernier Graphical Analysis Software with high accuracy (1mm).
Computer (x1)	Computer with a USB that allows the sensor to connect to the Vernier Graphical Analysis software.
Retort Stand (x1)	Must be rigid and have structural integrity.
Clamp & Rod (x1)	Must be rigid and have structural integrity

Table 2: Table of materials required for the experiment

*Note: while the spring constant is listed here, it was initially unknown and determined experimentally in the Preliminary Trials section.

1.7. Procedure

Setting up the lab

- (i) Place the retort stand on a flat surface and attach the rod to the retort stand using the clamp such that it holds itself tightly.
- (ii) Attach one end of the hook of the spring to the rod, while the other end should be attached to the mass hanger by inserting it through its hole.
- (iii) Add one slotted weight (20g) to the mass hanger.
- (iv) Place the ultrasonic motion sensor directly below the mass hanger.
- (v) Connect the ultrasonic motion sensor to the Vernier Graphical Analysis software via a computer and ensure that the sensor records the motion into a (new) position-time graph, setting the timer of the experiment to 120 seconds.
- (vi) Add 20g to the mass-hanger system.
- (vii) Bring the spring to its resting position by compressing it. Click "Record" and let go of the spring vertically, ensuring that the motion is as linear as possible, with little to no rocking.
- (viii) After 120 seconds, save the file and repeat step 7 five times to get five trials for one iteration.

- (ix) After completing five trials for an iteration, repeat steps 5 through 7 a total of five more times to get 6 iterations for masses of 40g, 60g, 80g, 100g, 120g, and 140g.

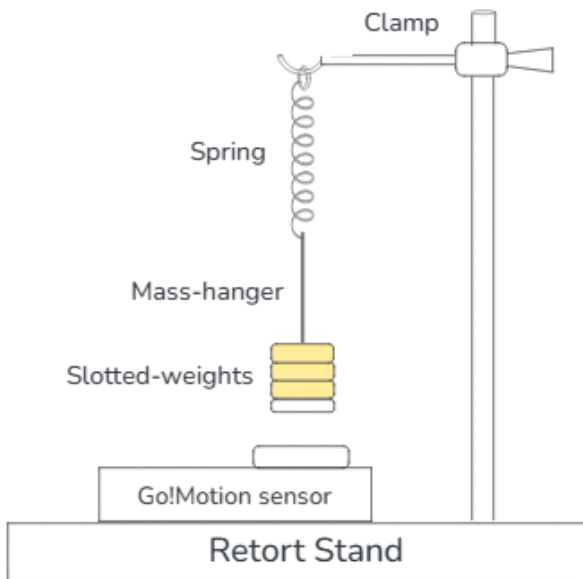


Figure 5: Apparatus diagram for the experiment

1.7.1. *Preliminary Trial: finding the k value*

To find the spring constant, we can use Hooke's law which states:

$$k = \frac{F}{x} \iff k = \frac{mg}{x}$$

Here, we ignore the typical negative sign present since we do not care about the direction of the motion. Now, to find the displacement of the spring (x), the following procedure was used:

- (i) To set up this preliminary trial, the same apparatus as the experiment was used, with the addition of a meter stick placed behind the spring.
- (ii) Firstly, the height of the spring without any mass was recorded using a meter stick. The rest height position was found relative to the clamp, with the uncertainty of this analog device recorded by dividing the smallest increment in half, which in this case was 1mm divided by 0.005m.
- (iii) Next, a mass m was added to the system and allowed to oscillate until there were no visible oscillations and a new equilibrium position was recorded.
- (iv) Upon reaching the equilibrium position, the equilibrium height was recorded relative to the clamp.
- (v) Finally, to find the spring coefficient Hooke's law was used, where x was calculated by using the net displacement by taking the difference of the rest and equilibrium position.

1.8. Risk Assessment and Safety

This experiment poses no risk as it does not jeopardize the safety of the experimenter or their surroundings, however just in case the experiment was carried out with utmost care to avoid blunders.

2. Data Collection

2.1. Qualitative Data

All the systems showed some damping, though the heavier ones seemed to show less damping, with almost all of them ending around the same final amplitude. However, it is to be noted that the motion was not entirely in 3D as the "rocking" or "swaying" motion of the mass-hanger and spring meant that there were irregularities while recording the oscillation (i.e., major spikes in the position-time graph).

2.2. Quantitative Data

This section includes all the data collected during the experiment.

	Height (m)	Uncertainty (m)
Initial	0.035	0.005
Final	0.135	0.005

Table 3: Initial and final height from preliminary trial using 3 slotted weights.

Time (s)	0.00	0.05	0.10	...	9.90	9.95	10.00
Position (m)	0.269	0.280	0.306	...	0.329	0.319	0.309

Table 4: Sample table of values collected during a trial (140g, fifth trial)

3. Data Analysis

3.1. Finding the damping ratio

3.1.1. *Finding the Instantaneous Damping Ratio*

The damping ratio was calculated by identifying oscillation peaks and applying the logarithmic decrement formula. The average of the results gave the mean logarithmic decrement, which was used to approximate the damping ratio as $\zeta = \delta/2\pi$. Below is the Python snippet for finding the peaks using an external library:

```
from scipy.signal import find_peaks # downloads external library
peaks, _ = find_peaks(position) # finds the peaks using external library
peak_amplitudes = position[peaks] # list of amplitudes at the peaks
```

Using this, the following table was constructed with the time and positions of all peaks within the dataset:

Time (s)	0.45	1.40	2.30	...	118.05	118.95	119.85
Position (m)	0.408520	0.408520	0.405070	...	0.140298	0.139113	0.137938

Table 5: Table of values with the time and positions at peaks (130 data points)

From this, the average logarithmic decrement was calculated as so:

```
# Make sure the data has at least 1 set of peaks
if len(peak_amplitudes) >= 2:
    # Find the damping coefficient (delta) for every successive peak
    deltas = np.log(peak_amplitudes[:-1] / peak_amplitudes[1:])
    delta_mean = np.mean(deltas) # compute the average of those peaks

    # Calculate damping ratio using the average damping coefficient
    damping_ratio = delta_mean / np.sqrt(4 * np.pi**2)
```

Below is a sample calculation for the fifth trial of 140g of weight at $t = 1.40\text{s}$, where $T = 0.90\text{s}$ (since adding 0.90s to 1.40s gives us the next successive peak at $t = 2.30\text{s}$):

$$\begin{aligned}
 \delta &= \ln \frac{x(1.40\text{s})}{x(1.40\text{s} + 0.90\text{s})} \\
 &= \ln \frac{0.41200\text{m}}{0.40852\text{m}} \\
 &= 0.00848247664 \text{ (3 s.f.g.)}
 \end{aligned}$$

Using this, the damping ratio is:

$$\zeta = \frac{\delta}{2\pi} = \frac{0.00848230016}{2\pi} = 0.00135$$

Using this, the following table was constructed, showing the damping ratio calculated for every mass for every trial:

	40g	60g	80g	100g	120g	140g
Trial 1	0.00287	0.0117	0.00164	0.00154	0.00342	0.00111
Trial 2	0.00296	0.00220	0.00170	0.00152	0.00155	0.00126
Trial 3	0.00282	0.00698	0.00163	0.00496	0.00159	0.00901
Trial 4	0.00295	0.00230	0.00172	0.00153	0.00154	0.00121
Trial 5	0.00519	0.00219	0.00170	0.00160	0.00142	0.00135

Table 6: Damping ratio for every mass for all 25 trials (3 s.f.g)

3.1.2. *Identifying outliers*

From this data, there were some visible outliers (i.e., 140g trial 3), however to calculate and identify any outliers (represented, the following formula was used:

$$x_n \text{ is an outlier if } \begin{cases} x_n < Q_1 - (1.5 * IQR) \\ x_n > Q_3 + (1.5 * IQR) \end{cases}$$

Where x_n represents a point of data for the n th trial. To find Q_1 , Q_3 , and IQR for a given increment, we use the following definition:

Q_1 = median of 25th percentile of data (first quartile)

Q_3 = median of 75th percentile of data (third quartile)

IQR = median of 50th percentile of data (second quartile)

For 140g for example, if we remove the third trial (x_3) as it is suspected to be an outlier, and calculate for Q_1 and Q_3 :

ordering the data set: [0.000918, 0.00110, 0.00132, 0.00160]

$$Q_1 = \frac{0.00111 + 0.00121}{2} = 0.00116$$

$$Q_3 = \frac{0.00126 + 0.00135}{2} = 0.001305$$

Since IQR is the second quartile, it can be calculated as:

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 0.001305 - 0.00116 \\ &= 0.000145 \end{aligned}$$

With this, we check if x_3 is an outlier.

- 1) $Q_1 - (1.5 * IQR) = 0.00111 - (1.5 * 0.000145) = 0.0008925$
- 2) $Q_3 + (1.5 * IQR) = 0.001305 + (1.5 * 0.000145) = 0.0015225$

$$\text{for } x_n \begin{cases} 0.00901 & \not> 0.0008925 \\ 0.00901 & > 0.0015225 \end{cases} \implies x_n \text{ is an outlier}$$

Therefore, it can be said with full certainty that x_3 is an outlier. Using this method, Table 7 can be revised as such to exclude the outliers:

	40g	60g	80g	100g	120g	140g
Trial 1	0.00287	—	0.00164	0.00154	—	0.00111
Trial 2	0.00296	0.00220	0.00170	0.00152	0.00155	0.00126
Trial 3	0.00282	—	0.00163	—	0.00159	—
Trial 4	0.00295	0.00230	0.00172	0.00153	0.00154	0.00121
Trial 5	—	0.00219	0.00170	0.00160	0.00142	0.00135

Table 7: Damping ratio for every mass for all 25 trials (3 s.f.g), without outliers

3.1.3. *Finding the Average Damping Ratio*

To calculate the average damping to compare it against the mass, the following formula was used:

$$\bar{x} = \frac{\text{sum of all trials}}{\text{number of trials}}$$

Below is a sample calculation for finding the average for 140g:

$$\begin{aligned} \bar{x} &= \frac{\text{sum of all trials}}{\text{number of trials}} \\ &= \frac{0.00111 + 0.00126 + 0.00121 + 0.00135}{4} \\ &= 0.00123 \end{aligned}$$

To find the absolute uncertainty of the damping ratios:

$$\Delta\zeta = \frac{\max \zeta - \min \zeta}{2}$$

Below is a sample calculation for 140g:

$$\begin{aligned} \Delta\zeta &= \frac{\max \zeta - \min \zeta}{2} \\ &= \frac{0.00135 - 0.00111}{2} \\ &= 0.0001 \text{ (1 s.f.g)} \end{aligned}$$

3.1.4. *Uncertainties*

To find the relative uncertainty, the following formula was used:

$$\delta\zeta = \frac{\Delta\zeta}{\zeta}$$

Below is a sample calculation for 140g:

$$\begin{aligned}\delta\zeta &= \frac{\Delta\zeta}{\zeta} \\ &= \frac{0.0001}{0.00123} \\ &= 0.08130 = 8.130\%\end{aligned}$$

With the average values and their uncertainties, the table for the mass vs the average damping ratio can be written as such:

Mass (g)	Δm	Damping ratio (ζ)	$\Delta\zeta$ (m)	$\delta\zeta$ (%)
40	0.8	0.002900	0.00007	2.414
60	2	0.002231	0.00006	2.689
80	2	0.001679	0.00004	2.382
100	2	0.001546	0.00004	2.587
120	3	0.001524	0.00008	5.249
140	3	0.00123	0.0001	8.130

Table 8: Masses vs Damping ratio (with absolute and relative uncertainty)

3.2. Linearization

As aforementioned, to check for a relationship between the mass and the damping ratio, the equation needs to be linearized. Using the following equation in the background section:

$$\frac{1}{\zeta^2} = \left(\frac{4k}{c^2} \right) m$$

It becomes evident that ζ needs to be transformed to $\frac{1}{\zeta^2}$. Doing so yields us the following values:

Mass (g)	ζ	$\rightarrow$	$\frac{1}{\zeta^2}$
40	0.002900	$\rightarrow$	118 906.064
60	0.002231	$\rightarrow$	200 909.679
80	0.001679	$\rightarrow$	354 730.563
100	0.001546	$\rightarrow$	418 389.735
120	0.001524	$\rightarrow$	430 556.417
140	0.00123	$\rightarrow$	660 982.22

Table 9: Converting damping ratio to its inverse square (for linearization)

3.2.1. *Uncertainties*

To calculate the uncertainties of $\frac{1}{\zeta^2}$, we let $y = \zeta^2$. We use equation (2) which defines how to calculate uncertainty for the form $y = \frac{ab}{c}$, and find the uncertainty for y as such:

$$\begin{aligned}
 &\text{given that } y = \frac{1}{\zeta \cdot \zeta} \\
 &\therefore \frac{\Delta y}{y} = \frac{\Delta \zeta}{\zeta} + \frac{\Delta \zeta}{\zeta} \\
 &\Delta y = y \left(\frac{2\Delta \zeta}{\zeta} \right) \\
 &\Delta y = \frac{1}{\zeta^2} \cdot \frac{2\Delta \zeta}{\zeta} \\
 &\Delta y = \frac{2\Delta \zeta}{\zeta^3}
 \end{aligned} \tag{1}$$

Below is a sample calculation using equation (1) for 140g:

$$\begin{aligned}
 \Delta y &= \frac{2\Delta \zeta}{\zeta^3} \\
 &= \frac{2(0.0001)}{0.00123^3} \\
 &= 107476.784 \\
 &= 200000 \text{ (rounded up to 1 s.f.g.)}
 \end{aligned}$$

To find the relative uncertainty, the same method as earlier was used, and Table 10 was revised as such¹:

Mass (g)	Δm	Damping Ratio (ζ)	$\Delta \zeta$ (m)	$\delta \zeta$ (%)
40	0.8	118 906	6 000	5.046
60	2	200 909	20 000	9.954
80	2	354 730	20 000	5.638
100	2	418 389	30 000	7.170
120	3	430 556	50 000	11.612
140	3	660 982	200 000	30.258

Table 10: Masses vs Inverse Squared Damping Ratio with absolute and relative uncertainties

3.2.2. *Graphing linearized data*

Using this linearized data, bringing $\zeta = \frac{c}{2\sqrt{km}}$ to the form of $y = mx + b$ by writing it as $\frac{1}{\zeta^2} = \left(\frac{4k}{c^2}\right)m$, we plot the graph as such:

3.3. *Approximating true slope*

To compare the accuracy of the slope of the graph $4k/c^2$, it is necessary to find the spring constant (k) and the damping coefficient (c).

¹ Noted that all the values below were rounded to 1 significant figure

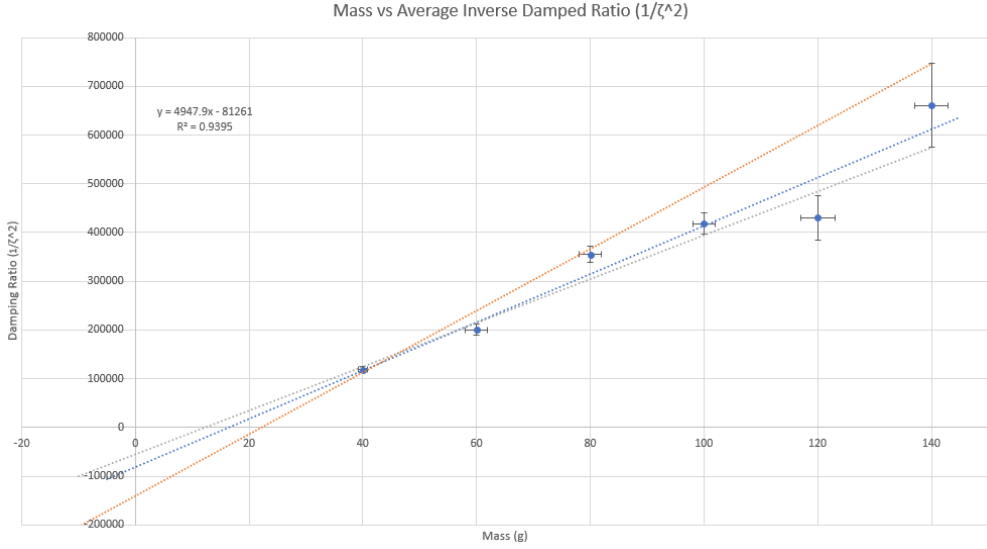


Figure 6: Mass vs Damping ratio (linearized)

3.3.1. Finding the spring coefficient

Using the experiment conducted in the Preliminary trials, it is clear that 3 slotted masses of $20 \pm 2\%$ g were used, which gave a final mass of 60.0 ± 1.2 g. The final mass (m_f) was found as such:

$$m_f = m_s + m_s + m_s = 0.20\text{g} + 0.20\text{g} + 0.20\text{g} = 0.60\text{g}$$

The absolute uncertainty of m_s is given by:

$$2\% * 20\text{g} = 0.4\text{g}$$

Using this, the absolute uncertainty of m_f was found as such:

$$\begin{aligned}\Delta m_f &= \Delta m_s + \Delta m_s + \Delta m_s \\ &= 0.4\text{g} + 0.4\text{g} + 0.04\text{g} \\ &= 1.2\text{g}\end{aligned}$$

The displacement, x was given by taking the final minus initial height:

$$\begin{aligned}x &= x_{\text{final}} - x_{\text{initial}} \\ &= 0.135\text{m} - 0.035\text{m} \\ &= 0.10\text{m}\end{aligned}$$

The uncertainty of which was:

$$\begin{aligned}\Delta(x) &= \Delta(x_{\text{final}}) + \Delta(x_{\text{initial}}) \\ &= 0.005\text{m} + 0.005\text{m} \\ &= 0.01\text{m}\end{aligned}$$

Using these values of mass and displacement, the spring coefficient was calculated to be:

$$k = \frac{mg}{x} = \frac{(60.0\text{g})(9.81\text{m/s}^2)}{0.10\text{m}} = 5886\text{g/s}^2$$

Given that:

$$\text{for } y = \frac{ab}{c}, \quad \frac{\Delta y}{y} = \frac{\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c} \quad (2)$$

The uncertainty can be found as such:

$$\begin{aligned} \frac{\Delta k}{k} &= \frac{\Delta m}{m} + \frac{\Delta g}{g} + \frac{\Delta x}{x} \\ \Delta k &= k \left(\frac{1.2\text{g}}{60.0\text{g}} + \frac{0\text{m/s}^2}{9.81\text{m/s}^2} + \frac{0.01\text{m}}{0.10\text{m}} \right) \\ &= (5886\text{g/s}^2)(0.12) \\ &= 706.32\text{g/s}^2 \\ &= 800\text{g/s}^2(1 \text{ s.f.g}) \end{aligned}$$

$\therefore$ The spring constant (k) = $5886 \pm 800\text{g/s}^2$

3.3.2. *Finding the damping coefficient*

While the true damping coefficient cannot be found (as mentioned earlier), it can be approximated using the equation of drag since the only resistive force is the drag faced by the cylinder in air, and can therefore be expressed as:

$$F_D = \frac{1}{2}\rho C_D A_C v^2$$

Where:

ρ – density of the fluid in which motion occurs (kg/m^3)

C_D – drag coefficient of an object (material's property, dimensionless)

A_C – cross-sectional area of object (m^2)

Since $F_D = -cv$, this gives the value of c as:

$$\begin{aligned} F_D &= \frac{1}{2}\rho C_D A_C v^2 \\ cv &= \frac{1}{2}\rho C_D A_C v^2 \\ c &= \frac{1}{2}\rho C_D A_C v \end{aligned}$$

Using the following values:

$$\rho = \text{density of air} = 1293\text{g/m}^3$$

$$C_D = \text{drag coefficient of short cylinder} = 1.15$$

$$A_C = \pi r^2 = \pi(0.032)^2 = 0.00322\text{m}^2$$

To find the value of v , the average drop velocity across all trials was used. This was found through the following Python code:

```
MASS = [40, 06, 80, 100, 120, 140] # all the masses used in the experiment
max_velocity = [] # list with the average velocity for each increment

# For every mass increment, calculate the average velocity
for increment in MASS:
    # loop through all 5 trials of the experiment
    for i in range(1, 6):
        trial_max = [] # list with the max velocity of each trial,
                        # reset every increment

        # In the case of outliers, do not use those files
        try:
            df = pd.read_csv(file_name)
            df.columns = ["Time", "Position", "Velocity", "Acceleration"]

            time = df["Time"].values
            velocity = df["Velocity"].values
            trial_max.append(max(velocity)) # append the max velocity for every
                                           # trial

            # We use max velocity since the sensor could start at different times,
            # and due to damping the max must be the first and the drop velocity

            # If there is an outlier, exclude it from the calculation
        except:
            pass

    max_velocity.append(np.mean(trial_max)) # find the average for every
                                           # increment and add it to a
                                           # list of averages for each
                                           # increment

print(np.mean(max_velocity)) # find the average of the drop velocity for
                             # every increment
```

A sample calculation (by hand would be as follows):

To find the average for all trials in an increment (in m/s):

$$x_{\text{avg}, 80} = \frac{\text{sum of all trials}}{\text{no. of trials}} = \frac{0.84485 + 0.85698 + 0.85270 + 0.80488 + 0.86114}{5} = 0.84411\text{m/s}$$

To find the average for all increments (in m/s):

$$x_{\text{avg}} = \frac{\text{sum of all trials}}{\text{no. of trials}} = \frac{0.84411 + 0.86716 + 0.86218 + 0.83636 + 0.89220}{5} = 0.8604\text{m/s}$$

The uncertainty of the velocity was found by using the data from the sensor¹, and calculated similarly to the $\Delta\zeta$:

$$\begin{aligned}\Delta v &= \frac{\max v - \min v}{2} \\ &= \frac{0.8620 - 0.8588}{2} \\ &= 0.0008 \text{ (1 s.f.g)}\end{aligned}$$

With this, the damping coefficient can be expressed:

$$\begin{aligned}c &= \frac{1}{2}\rho C_D A v \\ &= \frac{1}{2}(1293\text{g/m}^3)(1.15)(0.00322\text{m}^2)(0.8604\text{m/s}) \\ &= 2.0598\text{g/s}\end{aligned}$$

From this, the uncertainty of the damping coefficient can be found as such:

$$\begin{aligned}\frac{\Delta c}{c} &= \frac{\Delta v}{v} \\ \Delta c &= \frac{c \cdot \Delta v}{v} \\ &= \frac{2.0598\text{g/s} * 0.0008\text{m/s}}{0.8604\text{m/s}} \\ &= 0.002\text{g/s (1 s.f.g)}\end{aligned}$$

Using these values, the value for the slope and its uncertainty can be calculated. For the value of the slope (m_ζ):

$$\begin{aligned}m_\zeta &= \frac{4k}{c^2} \\ &= \frac{4(5886\text{g/s}^2)}{(2.060\text{g/s})^2} \\ &= 5549.19698\end{aligned}$$

¹ While not mentioned earlier, but the sensor included data on time, position, velocity and acceleration at any given time t .

The uncertainty of the slope can be found as such:

$$\begin{aligned}
 \frac{\Delta m_\zeta}{m_\zeta} &= \frac{\Delta k}{k} + \frac{\Delta c}{c} + \frac{\Delta c}{c} \\
 \Delta m_\zeta &= m_\zeta \left(\frac{\Delta k}{k} + \frac{2\Delta c}{c} \right) \\
 &= 5549.197 \left(\frac{800\text{g/s}^2}{5886\text{g/s}^2} + \frac{2(0.002)\text{g/s}}{2.060\text{g/s}} \right) \\
 &= 764.998313 \\
 &= 800 \text{ (1 s.f.g)}
 \end{aligned}$$

In conclusion, we find:

$$\begin{aligned}
 k &= 5886 \pm 800\text{g/s}^2 \\
 c &= 2.060 \pm 0.002\text{g/s} \\
 m_\zeta &= 5549 \pm 800
 \end{aligned}$$

3.3.3. Comparing slopes

To compare the expected slope (m_E) with the actual slope (m_A), we can use %error:

$$\begin{aligned}
 \%error &= \left| \frac{m_A - m_E}{m_E} \right| * 100\% \\
 &= \left| \frac{5549 - 4947.9}{4947.9} \right| * 100\% \\
 &= 12.15\% \text{ (4 s.f.g)}
 \end{aligned}$$

3.4. Spread of data

Using standard deviation alone can be misleading to judge the spread of data (specially with high-precision data where it may seem small), therefore the table below shows the coefficient of variation, which shows standard deviation relative to the mean:

Mass (g)	Standard Deviation (σ)	Coefficient of Variation ($\sigma/\bar{x}$)
40	0.00006904177611	2.240687818
60	0.00005712890008	2.842292048
80	0.00004108130740	2.239683732
100	0.00003815272353	2.334383390
120	0.00008404808870	4.819635935
140	0.00012052012020	8.203076779

Table 11: Standard deviation and Coefficient of Variation for each mass

4. Conclusion

Analysis of the data indicates that there is a positive correlation between the mass and the inverse square of the damping ratio, thus proving that as the mass increases, the damping ratio decreases, thereby confirming the hypothesis. However, the degree to which the data supports the hypothesis is somewhat questionable.

At first, the data appears to be accurate and precise with an R^2 value of 0.9395, indicating a strong correlation between the mass and the damping ratio. Furthermore, the precision is high due to the low relative uncertainties for the damping ratio ($< 10\%$) and a coefficient of variation of 2–8%, showing minimal spread in the data.

But, the uncertainties for the *inverse* damping ratio ($1/\zeta^2$) was much higher ranging from 5% to 30%, lowering confidence within the results. Moreover, the large number of outliers highlight the limitations of the sensor, as well as underlying systematic issues in the design due to (sometimes) poor data recording, bringing into question the reliability of the results. Furthermore, the lack of consistent sensor uncertainty also raises concerns about the reproducibility of the lab since each sensor has their own uncertainty depending on environmental and technical conditions.

Along with the precision, the accuracy of the experiment also falls in doubt as the y-intercept of the graph does not lie at the expected value of 0 at $m=0$, suggesting a systematic flaw in the design. Furthermore, the uncertainty of the trend line (max-min slopes) do not contain the origin, indicating that ultimately the investigation may have been plagued by uncontrolled factors, reducing trust in the results. However, despite the large %error of the trend line compared to the theoretical value ($>10\%$), the min and max values of the slope range include the expected value, suggesting that the experiment may have been successful to an extent.

In conclusion, while the experiment's reproducibility and (at times) accuracy is questionable, the results are acceptable and reaffirm the broader understanding of DHM systems.

5. Evaluation

Table 12: Strengths

Strength	Discussion
Large sample size	While the majority of the experiments use a sample size of 5×5 , this investigation utilized a 5×6 sample size, which only added confidence in the result and outlined a clearer trend.
Data processing	The method of data processing and analysis was a particularly strong point of this investigation, as it was conducted in great detail using various forms and types of data. Furthermore, the data analysis incorporated data processing, which helped in obtaining a more accurate result.
Methodology for finding the damping ratio	Recognizing the experimental limitations and finding an alternative method was a particularly strong point within the investigation, as it made examining the relationship between the mass and the damping ratio much easier.

Table 13: Weaknesses and Sources of Error in the Experiment

Weaknesses/ Sources of Error	Impact	Improvements
Lack of control on drag forces	One possible explanation for the outliers, as well as the general offset from the origin, could be the lack of control over the damping coefficient, particularly drag forces. While keeping the cross-sectional area constant might seem sufficient, the rocking motion caused when the object was released had a greater impact than expected, as the experiment was quite sensitive to change.	One possible improvement is to keep the volume of the apparatus the same but change the density to increase the mass. This would maintain aerodynamics while allowing for measurement of mass impact. Another approach could be to minimize drag forces by refining the object's shape.

Weaknesses/ Sources of Error	Impact	Improvements
The scale of the experiment	Although the experiment focused on underdamped motion, the changes were minimal and required a high degree of precision, which may not have been fully achieved (i.e., large uncertainties and y-intercept offset). The damping effect was not pronounced enough to be easily measured or visually observed without the ultrasonic sensor.	The damping effect could be exaggerated by modifying the design to incorporate an opposing force or increasing resistive forces (e.g., friction). Another alternative is to extend the trial duration to observe the damping effect more clearly over time.
Lack of accuracy in preliminary trials	The preliminary trials lacked validation since the experiment was not repeated multiple times to confirm accuracy, casting doubt on the reliability of the initial results.	Conduct the preliminary trial at least five times to ensure consistent and reliable results.
Different initial amplitudes	One uncontrolled factor was the initial amplitude of the drop, which varied across trials. The discrepancy arose from the delay between when the software began recording and when the mass was released. This inconsistency affected data collection.	To standardize the initial amplitude and eliminate human error, the process could be automated so that the sensor starts recording automatically when it detects the first oscillation.
Compression of the spring	While setting the spring to its rest position, unintended compression or extension may have occurred due to parallax error, making it difficult to determine its exact equilibrium state.	To control this, the length of the spring should be measured before each trial to ensure consistency.
Noise in the results	Some graphs, particularly those with outliers, showed noise and random spikes. This was likely due to the sensor recording measurements of background objects (e.g., distance from the clamp or ceiling).	A more precise ultrasonic sensor could be used, and the mass-hanger's motion should be controlled to ensure pure vertical oscillations without rocking or swaying.

Weaknesses/ Sources of Error	Impact	Improvements
Min-max uncertainty in ultrasonic sensor	The uncertainty in the ultrasonic sensor may have been greater than what was estimated using the max-min method, leading to uncertainty in the final results.	A better approach would be to calculate the standard deviation manually using preliminary trials and determine the standard error of the mean (SEM) to quantify the uncertainty more accurately.

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